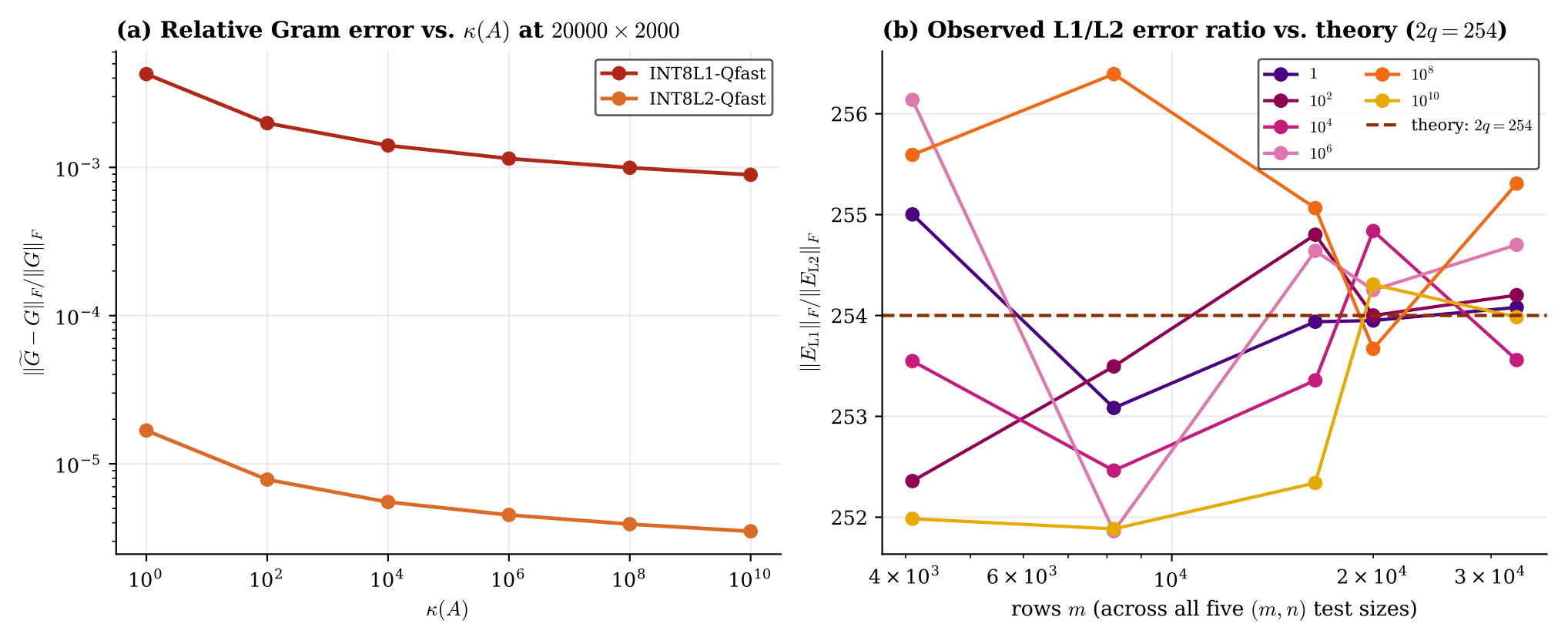
(a) Relative Gram error vs. $\kappa(A)$ at 20000×2000
INT8L1-Qfast
INT8L2-Qfast
$\|\widetilde{G} - G\|_F / \|G\|_F$
10^{-3}
10^{-4}
10^{-5}
10^0
10^2
10^4
10^6
10^8
10^{10}
$\kappa(A)$
(b) Observed L1/L2 error ratio vs. theory ($2q = 254$)
1
10^2
10^4
10^6
10^8
10^{10}
theory: $2q = 254$
$\|E_{L1}\|_F / \|E_{L2}\|_F$
256
255
254
253
252
4×10^3
6×10^3
10^4
2×10^4
3×10^4
rows m (across all five (m, n) test sizes)